

Swinburne University of Technology

COS30045 - Data Visualisation

Lab Exercise Demonstration 2 (Exercises 3.1 - 4.1)

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Lab 3.2

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<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8" />
  <meta name="description" content="Data Visualisation Exercise 1" />
  <meta name="keywords" content="HTML, CSS" />
  <meta name="author" content="Your name here" />

  <title>Task 3.2 D3 Drawing with Data - Scatter Plot</title>
  <!--Insert description of exercise -->
  <script src="https://d3js.org/d3.v7.min.js"></script>

</head>

<body>

  <h1>Creating a Scatter Plot with D3 Using Scales and Axis</h1>
  <script>
    // var dataset = [
    //   [5, 20], [500, 90], [250, 50], [100, 33],
    //   [330, 95], [410, 12], [475, 44], [25, 67],
    //   [85, 21], [220, 88], [550, 95], [600, 100]
    // ];

    var dataset = [];
    var numDataPoints = 20;
    var xRange = Math.random() * 1000;
    var yRange = Math.random() * 1000;

    for (var i = 0; i < numDataPoints; i++) {
      var newNumber1 = Math.floor(Math.random() * xRange);
      var newNumber2 = Math.floor(Math.random() * yRange);
      dataset.push([newNumber1, newNumber2]);
    }

    var w = 600;
    var h = 250;
    var padding = 40;
```

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var xScale = d3.scaleLinear()
  .domain([0, d3.max(dataset, function (d) {
    return d[0];
  })]) // The data input range
  .range([padding, w - padding]); // The range available for
visualization on screen

var yScale = d3.scaleLinear()
  .domain([0, d3.max(dataset, function (d) {
    return d[1];
  })]) // The data input range
  .range([h - padding, padding]); // The range available for
visualization on screen

var xAxis = d3.axisBottom().ticks(5).scale(xScale);
var yAxis = d3.axisLeft().ticks(5).scale(yScale);

var svg = d3.select("body")
  .append("svg")
  .attr("width", w)
  .attr("height", h);

svg.selectAll("circle")
  .data(dataset)
  .enter()
  .append("circle")
  .attr("cx", function (d, i) {
    return xScale(d[0]);
  })
  .attr("cy", function (d) {
    return yScale(d[1]);
  })
  .attr("r", 5)
  // .attr("fill", "slategrey");
  .attr("fill", function (d) {
    // Conditional to make points red if x > 300 or another condition
    you like
    if ((d[0] > 480)) {
      return "blue"; // Red color for some points
    } else {
      return "slategrey"; // Default color
    }
  });

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svg.selectAll("text")
  .data(dataset)
  .enter()
  .append("text")
  .text(function (d) {
    return d[0] + "," + d[1];
  })
  .attr("x", function (d) { return xScale(d[0]); })
  .attr("y", function (d) {
    return yScale(d[1] + 3);
  })
  .attr("fill", "red")
  .attr("font-family", "sans-serif")
  .attr("font-size", "11px")

svg.append("g").attr("class", "x axis").attr("transform", "translate (0,
"+ (h - padding) + ")").call(xAxis).selectAll("text")
  .style("text-anchor", "center")
  .style("font-size", "12px")
  .style("fill", "blue");

svg.select(".x.axis")
  .select(".domain")
  .attr("stroke", "blue");

svg.select(".x.axis")
  .selectAll(".tick line")
  .attr("stroke", "blue");

svg.append("g").attr("class", "y axis").attr("transform", "translate(" +
padding + ",0)").call(yAxis).selectAll("text")
  .style("text-anchor", "center")
  .style("font-size", "12px")
  .style("fill", "blue");

svg.select(".y.axis")
  .select(".domain")
  .attr("stroke", "blue");

svg.select(".y.axis")

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        .selectAll(".tick line")
        .attr("stroke", "blue");

</script>

<br>
<footer style="color:grey">COS30045 Data Visualisation<br>Joe Bloggs</footer>

</body>

</html>

```

